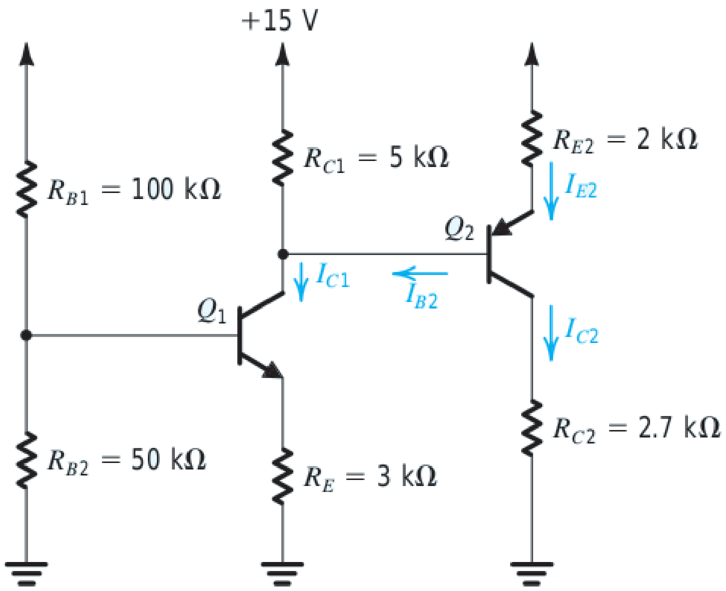
BJT Circuits at DC

Example 6.11

- We want to analyze the circuit in Fig. 6.29(a) to determine the voltages at all nodes and the currents through all branches. Assume $\beta = 100$ for both transistors.



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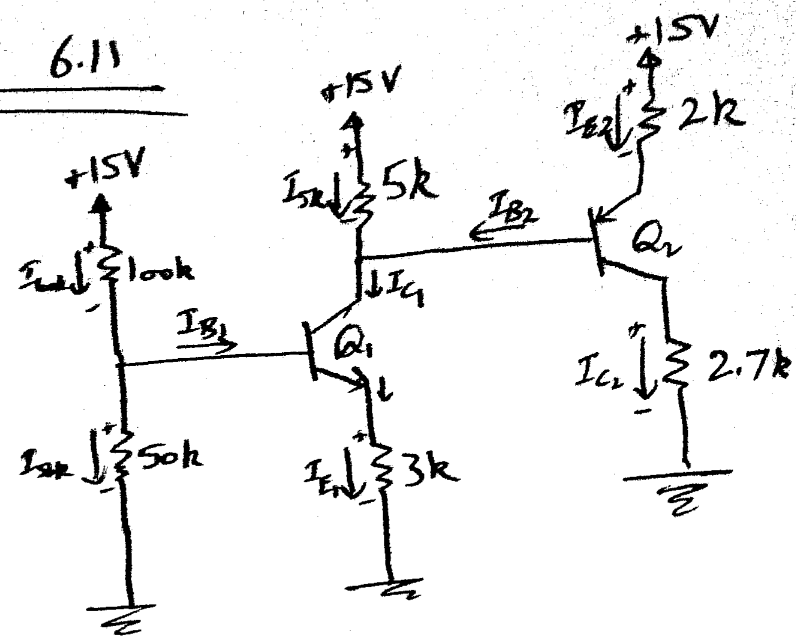
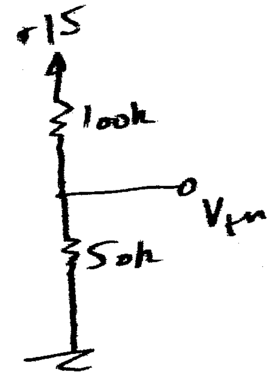
Example 6.11



Q_1 is npn, Q_2 is pnp



finding Thevenin Equivalent

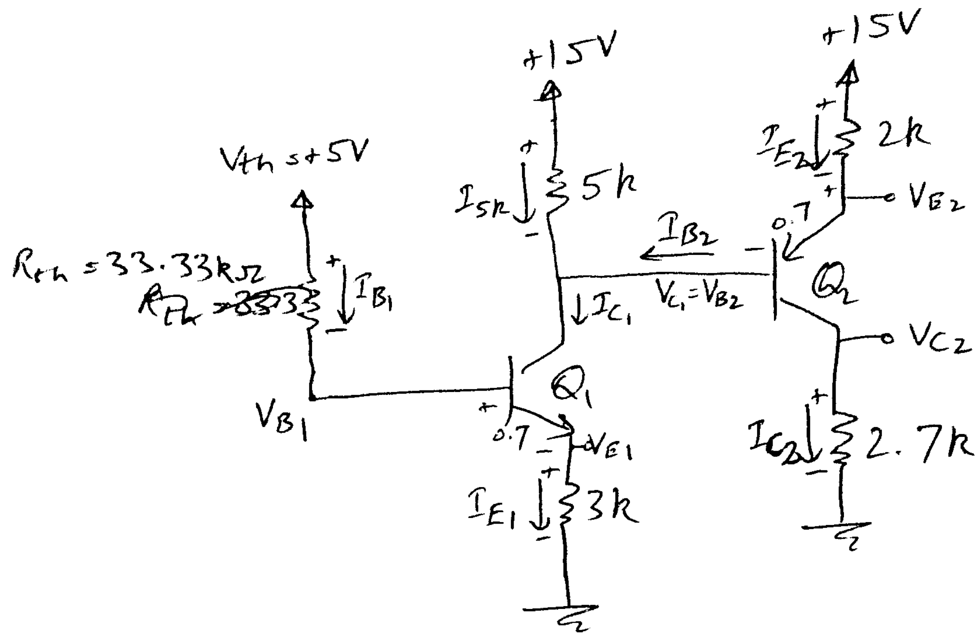
$\Rightarrow R_{th} = 100k // 50k$

$= 33.33k$

by voltage divider

$$V_{th} = \frac{50k}{50k + 100k} 15V$$

$= 5V$



Assuming both Q_1, Q_2 are in active mode

here $\beta = \beta_1 = \beta_2 = 100$, $\alpha = \alpha_1 = \alpha_2 = \dfrac{\beta}{\beta+1}$

$$= \frac{100}{101} = 0.9901$$

KVL for Q_1

$$\Rightarrow 5 = (33.33k) I_{B_1} + 0.7 + (3k) I_{E_1}$$

$$\because I_{E_1} = I_{C_1} + I_{B_1} = \beta I_{B_1} + I_{B_1} = (\beta+1) I_{B_1}$$

$$5 = (33.33k) I_{B_1} + 0.7 + (3k)(\beta+1) I_{B_1}$$

$$5 - 0.7 = \{33.33k + 3k(\beta+1)\} I_{B_1}$$

$$4.3 = (33.33k + 3k(101)) I_{B_1}$$

$$I_{B_1} = \frac{4.3}{336.33k} = 0.0128\,mA$$

$$I_{E_1} = (\beta+1) I_{B_1} = (101)(0.0128m) = 1.293\,mA$$

$$I_{C_1} = \beta I_{B_1} = (100)(0.0128m) = 1.28\,mA$$

By ohms' law

$V_{E_1} = (3k)\, I_{E_1} = (3k)(1.293m) = 3.879V$

By KVL

$V_{B_1} = 0.7 + V_{E_1} = 0.7 + 3.879 = 4.579V$

KCL $\Rightarrow$ $I_{5k} + I_{B_2} = I_{C_1}$

$\boxed{I_{5k} = I_{C_1} - I_{B_2} = 1.28m - I_{B_2}} \rightarrow (1)$

by ohms law

$15 - V_{C_1} = (5k)\, I_{5k} \longrightarrow (2)$

$15 - (5k) I_{5k} = V_{C_1}$

also $V_{C_1} = V_{B_2}$

Now for Q_2

Assume Active mode

KVL for Q_2

$\Rightarrow 15 = (2k)\, I_{E_2} + 0.7 + V_{B_2}$

$15 - 0.7 = (2k)\, I_{E_2} + V_{C_1}$ $\quad \because V_{B_2} = V_{C_1}$

$14.3 = (2k) I_{E_2} + \left[15 - (5k) I_{5k}\right]$ $\quad \because V_{C_1} = 15 - (5k) I_{5k}$ because of (2)

$14.3 = (2k) I_{E_2} + 15 - (5k)(1.28m - I_{B_2})$ $\quad \because$ of (1)

as $I_{E_2} = (\beta + 1) I_{B_2} = 101\, I_{B_2}$

$\Rightarrow 14.3 = (2k)(101) I_{B_2} + 15 - (5k)(1.28m) + (5k)\, I_{B_2}$

$14.3 - 15 = (202k) I_{B_2} - 6.4 + (5k) I_{B_2}$

$14.3 - 15 + 6.4 = (202k + 5k) I_{B_2}$

4/5

$$5.7 = (207k) I_{B_2}$$

$$I_{B_2} = \frac{5.7}{207k} = 0.02754\ mA$$

$$I_{E_2} = (\beta+1) I_{B_2} = (101)(0.02754m) = 2.7815mA$$

$$I_{C_2} = \beta I_{B_2} = (100)(0.02754m) = 2.754mA$$

Eq. 1 $\Rightarrow$ $I_{5k} = 1.28m - I_{B_2}$

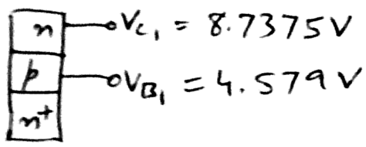
$= 1.28m - 0.02754m$

$= 1.2525mA$

Eq 2 $\Rightarrow$ $V_{C_1} = 15 - (5k)(1.2525m)$

$= 8.7375V$

Thus for Q_1



$V_{C_1} > V_{B_1}$

$\Rightarrow$ CBJ of Q_1 is reverse biased

$\Rightarrow$ Q_1 is indeed in active mode.

for Q_2

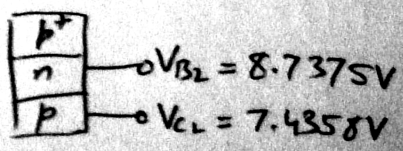
by ohms' law

$V_{C_2} = (I_{C_2})(2.7k) = (2.754m)(2.7k)$

$= 7.4358V$

$V_{B_2} = V_{C_1} = 8.7375V$

Thus for Q_2



here $V_{B_2} > V_{C_2}$

$\Rightarrow$ CBJ of Q_2 is reversebiased $\Rightarrow$ Q_2 is indeed in active mode

BJT Circuits at DC

- by KVL
  - $V_{E2} = V_{B2} + 0.7 = 8.7375 + 0.7 = 9.4375V$
- by ohm's law
  - $I_{50k} = \frac{V_{B1} - 0}{50k} = \frac{4.579}{50k} = 0.0916mA$
- by ohm's law
  - $I_{100k} = \frac{15 - V_{B1}}{100k} = \frac{15 - 4.579}{100k} = 0.10421mA$